

Radon Monitoring - User Manual

Version 4.2.1 | Home Assistant App for GQ RadonScan devices

Local • Read-only • Scientific analysis • Controlled data management

Radon Monitoring

Radon Monitoring reads completed hourly measurements from the local SPIR history of compatible GQ RadonScan devices using read-only communication. It stores data locally in SQLite, publishes Home Assistant entities and provides analysis, scientific reports, World Map upload and controlled data management.

Home Assistant App • Version 4.2.1

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1. Introduction

Radon Monitoring reads completed hourly measurements from the local SPIR history of compatible GQ RadonScan devices using read-only communication. It stores data locally in SQLite, publishes Home Assistant entities and provides analysis, scientific reports, World Map upload and controlled data management.

2. Installation and first start

- Connect the GQ RadonScan to the Home Assistant host by USB.
- Provide an MQTT service for apps.
- Install and start Radon Monitoring.
- Leave serial_port empty for automatic detection, or enter a stable device path.
- Open the Radon Monitoring panel and verify the device status.

3. User interface

The sidebar separates tasks deliberately: Overview for daily operation, Analysis for scientific evaluation, Expert view for protocol and diagnostics, History for tables and exports, Reports for PDF creation, Data management for backup and deletion, and GQ World Map for external uploads. On mobile devices the sidebar becomes a slide-out menu.

4. Measurements and data quality

24-hour, 7-day and 30-day values are displayed only after the corresponding period has elapsed and the configured minimum coverage has been reached. Missing hours are not replaced. Instead of a misleading number, the tile explains why a result is not yet meaningful.

5. Analysis and statistics

Analysis includes arithmetic and geometric means, median, standard deviation, percentiles, interquartile range, median absolute deviation, robust Theil-Sen trends, threshold duration, data gaps, daily and weekly profiles and event markers. Time-based profiles use the configured analysis time zone while storage remains in UTC.

6. GQ Radiation World Map

GQ Radiation World Map upload is disabled by default. After activation, account and device IDs are read from app configuration. A persistent queue prevents duplicates and retries temporary failures. The app does not transmit GPS coordinates; the public position is maintained in the GQ device profile. Review the privacy implications before enabling external uploads.

7. Reports

Compact and detailed scientific PDF reports document the period, device, measurement site, coverage, statistics, charts, events, conversion factor, software version, methodological limitations and a SHA-256 checksum of the selected canonical data. Reports do not replace official, medical or building-professional assessment.

8. Data management and restore

Data management can create complete ZIP backups, validate and restore SQLite databases and delete measurements by period, device or campaign. The app creates an automatic backup before destructive local operations. Physical device history is never modified. Home Assistant Recorder history is managed separately and only for explicitly selected entities.

9. Home Assistant integration

MQTT Discovery retains the established version-3 identifiers, avoiding duplicate entities during upgrade. The app can emit Home Assistant events for new campaigns and World Map uploads. Because it includes deletion and restore operations, the panel is restricted to administrators.

10. Configuration

Option	Default	Purpose
scan_interval	300 s	Interval between read-only device polls
factor_bq_m3_per_cph	1.530	Conversion factor stored with each measurement
minimum_data_coverage_percent	95	Required coverage for period values
history_retention_days	1095	Retention of local measurements
analysis_timezone	auto	Time zone for daily and weekly profiles
gmcmmap_upload_interval_minutes	60	Automatic World Map upload interval
gmcmmap_max_age_hours	72	Maximum age for queued automatic uploads
gmcmmap_retry_limit	8	Maximum number of upload attempts

11. Troubleshooting

- No device: check USB, permissions and serial_port; ensure no other program holds the port.
- No new values: only completed hours are published; also check data age and Expert view.
- Average unavailable: the required duration or coverage has not yet been reached.
- World Map error: verify account ID, device ID, internet access and queue status.
- Report failure: check free storage and database integrity.
- After upgrade: restart once and verify successful database migration in the log.

12. Safety and scientific limitations

The app uses read-only device commands and does not modify memory or configuration. SPIR decoding is supported by measurement series but is not an official GQ protocol specification. Short-term values must not be interpreted as a statutory annual mean. Health, workplace or building decisions require appropriate long-term measurements and, where necessary, qualified professionals.

Technical note

Database path: /data/radonscan_v3.sqlite3. Internal app slug: gq_radonscan. Device communication remains read-only. The separate bundled protocol reference documents the confirmed SPIR read path and is not an official manufacturer specification.